STEP-BY-STEP REPAIR PRIORITY

Do the work in this order

Skipping ahead replaces parts on a truck that may still be a fire risk, and it teaches the PCM a leak. Each line is the action. The note under it is why, and how sure we are.

1. **STOP DRIVING** if the manifold is glowing or oil is reaching hot exhaust. STOP
   Active orange or red iron, or oil on the pipe or a sensor, is a safety stop. Guides 05 and 06. Do not idle it to confirm.
2. **Let the vehicle cool completely.**
   An hour is a minimum after a glow. Do not spray water on the manifold. Do not degrease a hot pipe. Solvent on hot iron is a fire.
3. **Find and fix the vacuum leak.** TEST REQUIRED
   Smoke, visual, and the brittle CCV, EVAP, booster, and intake joints in Guide 04. A leak is a LIKELY cause of high idle on this MAP engine. It is only a POSSIBLE cause of a lean glow. Fix what you find. Do not relearn yet.
4. **Replace the damaged IAC with a NEW correct unit.** CATALOG LISTED
   The pintle was moved. Treat the motor as damaged even if the coils ohm in range. Mopar 4874373AB is the catalog candidate for a 1998–2004 Grand Cherokee. Verify by VIN. No used motors.
5. **Do NOT move the new IAC pintle.** RULE
   Not in the box, not to “seat” it, not to clean behind the tip. The PCM counts steps. A pushed tip makes the count a lie. If the new one does not sit flush, look for a pinched O-ring. Do not force the pintle.
6. **Clean the throttle body.**
   Plate edge and bore, key off. The plate must close and the cable must return. A cable that holds the plate open will not be fixed by any IAC. Do not soak the TPS.
7. **Reset the PCM and perform the idle relearn.**
   Only after steps 3 through 6. Battery negative off 15 minutes, or a scan tool that clears adaptive memory. Key on 10 seconds, no throttle, then the warm idle procedure in Guide 03. This is a cable-throttle engine. There is no pedal relearn.

REPAIR PRIORITY · FINISH AND PROVE

8 Verify warm idle. TEST REQUIRED

Fully warm, Park, loads off. Screening band about 700–800 rpm. TSB reference about 744 rpm on the cited 2004 calibration. Record RPM and coolant temperature. Above about 1,000 rpm means go back to the leak, the plate, and the IAC O-ring. Do not reset again and hope.

9 Locate and repair the oil leak. TEST REQUIRED

Degrease cold. Watch fresh oil. Common 4.0L sources: valve cover, filter-adapter O-rings, oil-pressure sender, pan gasket, rear main. LIKELY places. The fresh track decides. Do not replace a rear main until the adapter and the pan are ruled out.

10 Recheck and repair the P0161 heater circuit. TEST REQUIRED

Separate fault. Bank 2 Sensor 2, rear downstream, after the rear mini-cat. Connector, fuse, cold heater ohms (screen 4–10 Ω), 10–12 V while the ASD is actually on, then PCM control. Replace the sensor only if the element or the sensor connector failed, and only after oil is no longer reaching it.

11 Clear codes.

Only after the test record is filled and the repair those tests support is done. Freeze frame is gone once you clear. Photograph the scan tool first.

12 Start, monitor, and rescan.

Idle until warm. Watch the manifold from the side. Watch for a new oil track. Read RPM. Rescan. A code that returns on this start is still a real fault.

13 If the manifold glows again, STOP immediately. STOP

Key off. Cool it. Tow it. Do not repeat the relearn. Go back to Guide 05: lean condition, misfire, timing signal, and a restricted catalyst are all still open until tested.

Finished only when all of these are true

The manifold stays dark. No fresh oil reaches the pipe or a sensor. Warm idle is in the 700–800 rpm screen in Park with loads off. P0505 and P0161 stay off after the drive cycle you can safely complete. Anything less is an unfinished test, not a comeback mystery.

The order above is the repair priority for this truck's stated history: a pintle that was moved, a high-idle / P0505 concern, a possible lean-heat path, a separate oil leak, and a separate P0161. It is not a claim that every arrow in that history has been measured.